

AI in Healthcare

Hype · Hope · Evidence

“Where is the beef?”

θ theory · p real-world · τ toy

Seymour Papert’s three classes of AI problems.

We take our name from p — we exist to solve the hard, real-world problems, grounded in θ theory.

Four builds across four decades



B.Tech · IIT Madras

Ph.D. · Buffalo (SUNY)

S.M. · MIT Sloan



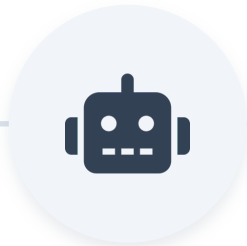
Oracle

DB Java internals



HP

Web services · E-Speak
Tech Strategy for Services



DxContinuum → ServiceNow

AI platform · \$0→\$250M



ThetaRho

AI in healthcare

Hype → Hope → Evidence → Open

01



HYPE

what's happened

02



HOPE

where AI helps

03



EVIDENCE

data & truth

04



OPEN

what's unsolved

Payoff at the end: a checklist for evaluating any AI solution.

The hype has always outrun reality



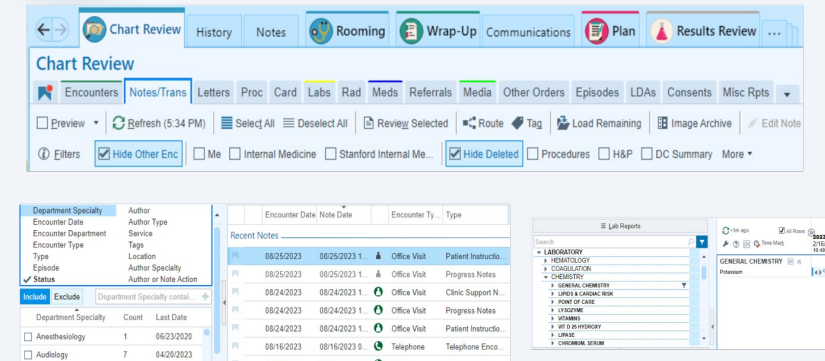
The vision

“80% of a physician’s work” — a VC, 2012

IBM Watson Health · Google Health · record investment



The reality



40% see more EHR burden than benefit

“*Vision without execution is hallucination.*” — Thomas Edison

A tale of two cities



Clinical AI

GETS THE PRESS

1,500+ FDA-cleared algorithms

narrow by design · one per condition



Administrative AI

THE REAL PAIN

2,541 clicks in a day of visits

spans every specialty · compounds



“I feel like a data-entry clerk, not a physician.”

The ask: help me — don’t replace me. In both cities, the user is an afterthought.

Administrative AI: front, mid, back office



FRONT

Access & scheduling

SAMPLE COMPANIES

Hippocratic, Sully, Sierra



MID

Clinical support

SAMPLE COMPANIES

**Ambience, Heidi,
OpenEvidence**



BACK

Revenue cycle & prior auth

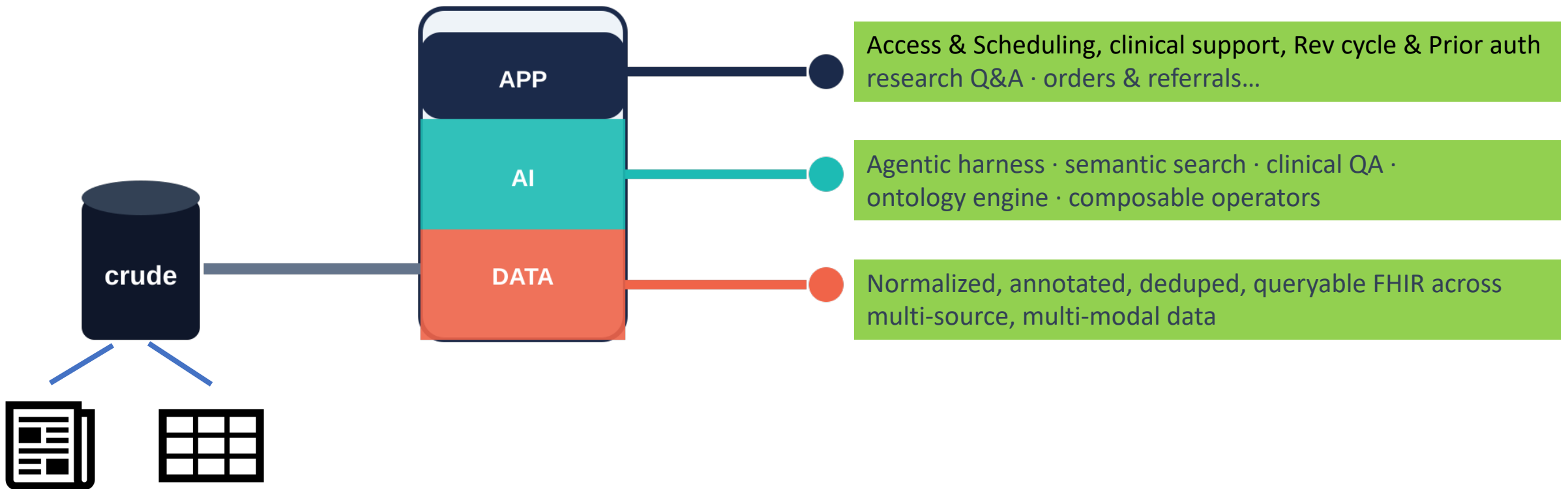
SAMPLE COMPANIES

**Automation Anywhere,
UiPath**

Different app per office — same grounded data layer underneath. That's the wedge.

Healthcare data is crude oil. AI needs jet fuel. We built the refinery.

Refine → Reason → Act



Patient data lives in many places — and arrives messy

EHRs, HIEs, claims, devices, patient-generated data — multi-modal, duplicated, inconsistently coded

EHR systems

Epic, Cerner, athena, Meditech — different schemas, different code sets

HIEs / networks

CommonWell, Carequality, TEFCA — varying conformance, lossy hand-offs

Claims

ICD-10-CM, CPT-4 — billing-shaped, not clinically shaped

Labs & devices

Mixed LOINC adoption, vendor-specific local codes

Notes & PDFs

Free text, scanned documents, dictated narrative

Patient-generated

Wearables, apps, questionnaires — outside the EHR boundary

WHAT ACTUALLY ARRIVES

A noisy, partially-coded blob

Duplicates across feeds

Same concept, different codes

Codes missing or stale (ICD-9 in 2026)

Free-text where there should be codes

Codes where there should be free text

Conflicting timelines & dates

PDFs masquerading as structured data

FHIR is what makes any of this tractable

One resource graph instead of N vendor schemas — adapting per-vendor would be impractical

Without FHIR

$N \times M$

integrations — every consumer adapting to every vendor's data model

With FHIR

$N + M$

each producer normalizes once; each consumer reads one shape

WHAT FHIR GIVES YOU

A shared data model for everything clinical

Resources

Condition · Procedure · Medication · Observation · DiagnosticReport · DocumentReference · Encounter · ...

CodeableConcept

Every clinical fact carries (system, code, display) — pluggable to any vocabulary

Identity & links

Patient · Encounter · Practitioner stitching across resources

Provenance

Where each fact came from, when, asserted by whom

Bulk & search

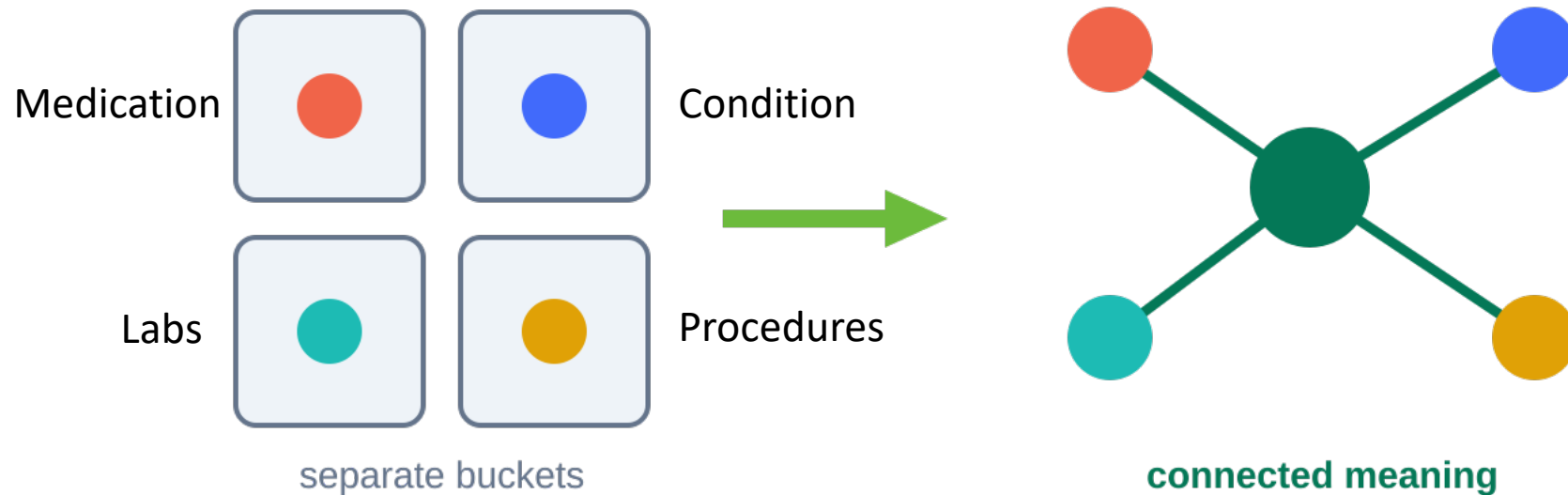
\$everything, _include, _revinclude — the substrate is queryable

Raw data → one grounded FHIR store



Plugin per source · OCR for PDFs & images · dedup verdict · Person map · multi-tenant · full lineage

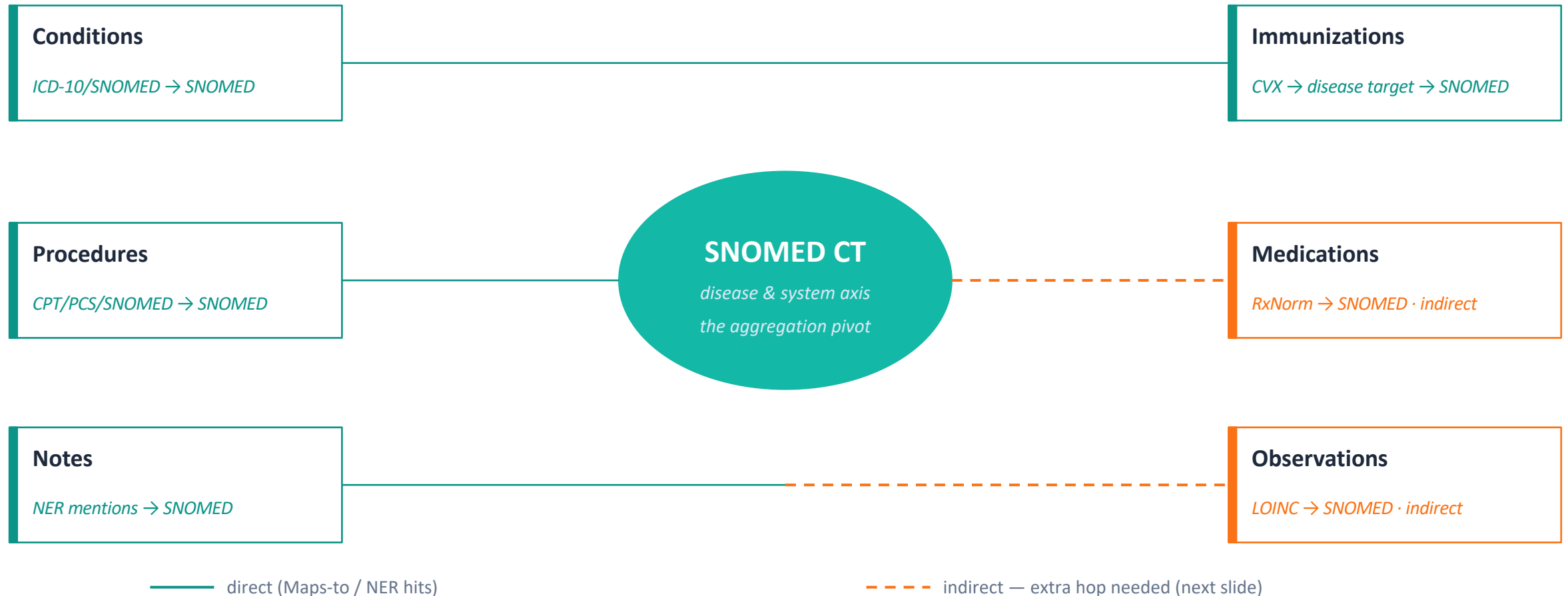
FHIR gives you shape. Ontology gives you meaning.



FHIR can't say metformin treats diabetes. The ontology can — for query, and for grounding.

Everything needs to land in SNOMED, as it is clinical vocabulary

RxNorm and LOINC are necessary — but not sufficient — for the user's question



Medications and labs don't sit in SNOMED — but their orders do

An order carries an indication: a Condition that does sit in SNOMED

MEDICATION → SNOMED

- 1 MedicationRequest in RxNorm**
metformin · RxCUI 6809
- 2 Linked MedicationRequest.reasonCode / reasonReference**
the indication recorded at order time
- 3 Condition: Type 2 diabetes mellitus**
ICD-10 E11.9 — already SNOMED-mappable
- 4 Maps-to → SNOMED 44054006**
now the medication is anchored to a disease
- 5 Walk to body system**
diabetes → endocrine/metabolic axis

OBSERVATION (LAB) → SNOMED

- 1 DiagnosticReport / Observation in LOINC**
HbA1c · LOINC 4548-4
- 2 Linked ServiceRequest.reasonCode**
why the lab was ordered
- 3 Condition: Type 2 diabetes mellitus**
or screening / monitoring rationale
- 4 Maps-to → SNOMED**
lab is now anchored to the disease
- 5 Walk to body system**
endocrine/metabolic — same axis

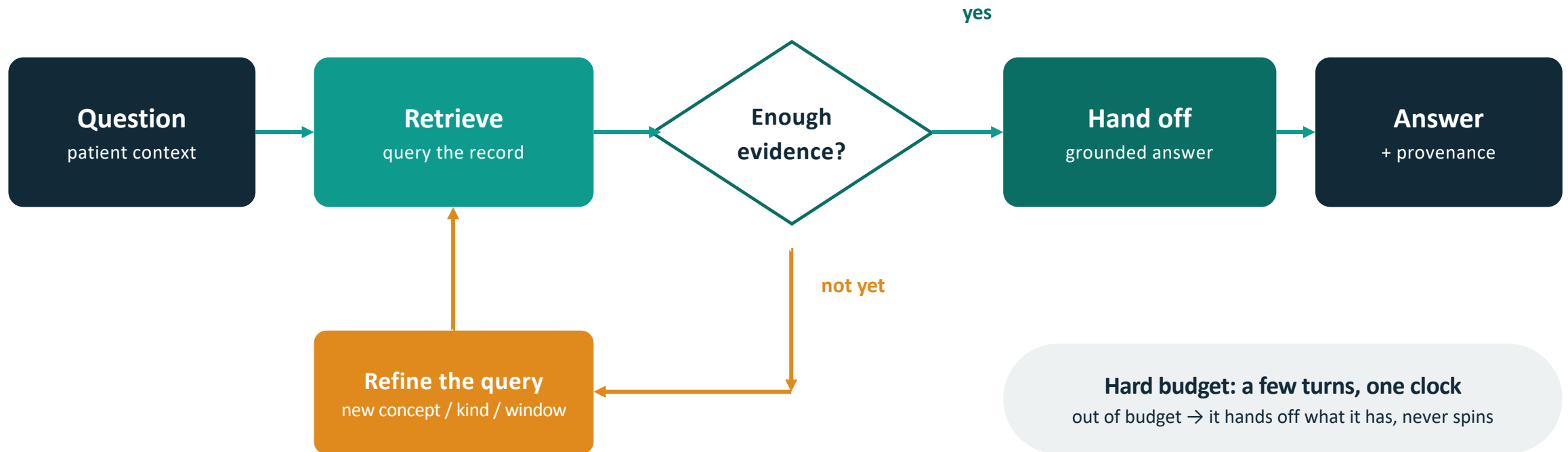
One context: retrieve, research, act



Every output grounded in FHIR · hallucinations flagged · full audit trail.

A minimal self-correcting loop

A single retrieval can come back confidently wrong, with no signal that it missed. The loop adds that missing feedback.



If the loop ever stalls, errors, or comes back empty, it falls back to the standard one-shot result — so turning it on can never make a question fail.

Built to stay in bounds

A free-running loop can wander or repeat itself. Every recovery move here is counted and capped — none is an open-ended retry.

Always terminates

A hard cap on turns and a wall clock. When either runs out, the loop stops and hands off the evidence it already has.

A nudge, not a spiral

When the evidence looks thin, the loop gets exactly one prompt to query differently — counted, so it can't loop on the hint.

Self-repairing calls

Malformed tool arguments are deterministically repaired — numbers, booleans, enum casing — before anything is rejected. No guessing.

No repeat queries

A repeat detector catches the same query twice and asks for a genuinely different angle instead of re-fetching identical rows.

Fail-safe by design

Any failure falls back to one-shot retrieval. The loop can be enabled without ever becoming a new way for an answer to break.

Privacy-safe telemetry

Every step is traced with structure only — tool names, counts, timings — and no clinical content, so runs are fully auditable.

The result: the reach of an agent that can gather evidence over several steps, with the predictability of a bounded pipeline.

Grounding under every step



Domain ontology sits under every step — annotate, retrieve, then compare answer to source code-to-code to catch fabricated facts.

Hold the AI to a courtroom standard



The truth

no fabrication

The whole truth

no omission

Nothing but the truth

no irrelevance

Four honest states



GREEN

validated



RED

contradiction



YELLOW

candidate hallucination

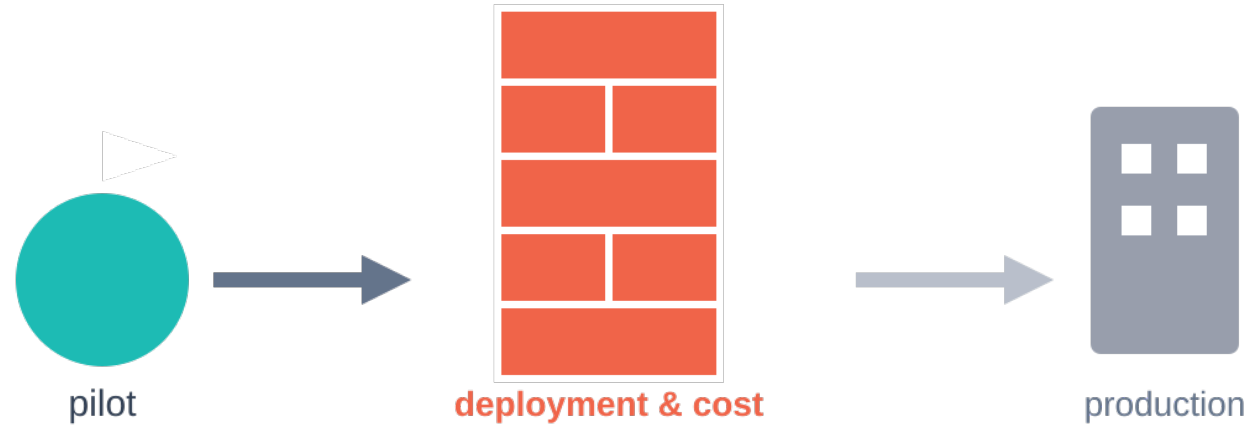


GRAY

not checked

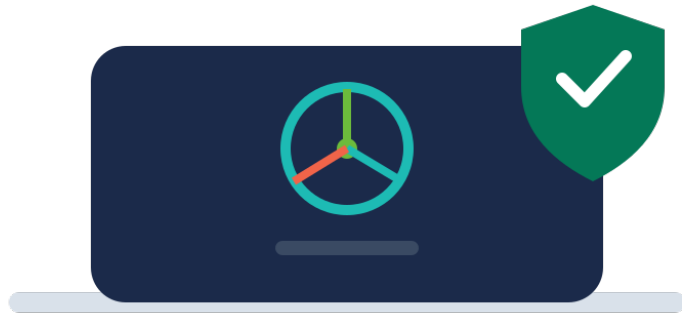
“Not checked” must never look like “trusted.” GRAY makes the uncovered part of every answer visible.

Functionality gets the pilot. Production is another matter



Two walls: the pilot-to-production gap, and cost that scales with usage — plus data sovereignty.

The appliance: sovereign, predictable, on-device



- **Data never leaves the building**
“trust us” → “you control it”
- **Fixed capex, not per-token opex**
scale at your own pace
- **~30B param open models, local inference**
enough for scoped, grounded tasks
- **Governed by the customer’s IT**
their policies, their hardware

Compliance by design — and tools, via MCP



Compliance by design

Partitioned isolation · immutable audit · data-minimization ·
HIPAA & SOC 2



Exposed as tools · MCP

9 healthcare tools · 5 domains. Any MCP agent can retrieve,
reason, act — guardrails intact.

“Nothing new under the sun” — the web-services idea I built at HP, now pointed at agents.

What to ask of any AI solution



Grounding & provenance



Honesty about coverage



Completeness where it matters



Data before model

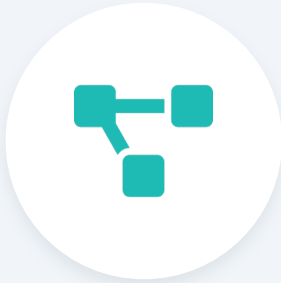


Adoption evidence



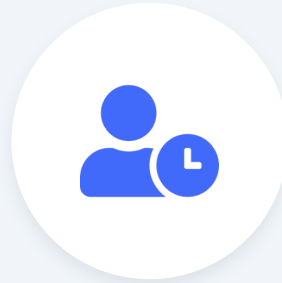
Economics & sovereignty

Where the cutting-edge research is heading



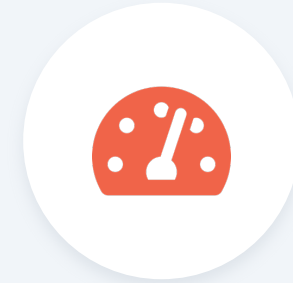
Reasoning you can trust

verify every step



Models of the whole patient

longitudinal · causal



Safe · calibrated · measurable

know when not to answer

The base model is no longer the bottleneck — the work moved up the stack.

IN CLOSING

Hype

the user was an afterthought

Hope

grounded, longitudinal data

Evidence

the truth, the whole truth, nothing but

The beef is real — at the intersection of what's economically forced and technically feasible. Users are the bridge.



Thank you. kannan@thetarho.ai